



Standardized use inspection of workers' personal protective equipment based on deep learning

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ABSTRACT

Falling from height (FFH) and object strike (OS) accidents frequently occur at construction sites, threatening personnel safety and affecting the construction quality. In the hierarchy of controls, personal protective equipment (PPE) is the most easily achievable measure. However, the presence of hazards and the differentiation in subjective protection awareness make PPE the least effective control measure. Therefore, it is necessary to promote proper and standardized use of PPE to meet the requirements from the perspective of administrative control. Taking two behaviors that can lead to OS and FFH accidents as a research case—loosening the hardhat and not using the safety harness's hook—this study proposes a deep-learning-based inspection method. First, we established a detection model for hardhats and hooks based on You Only Look Once v5. Thereafter, the object-detection model and Openpose algorithm were applied to recognize 1200 video clips containing three unsafe behaviors and one safe behavior and generate 1200 data files that vary in a time series. Finally, a one-dimensional convolutional neural network (1D-CNN) model was trained with the data of 600 videos, and the model was used to test the data of the other 600 videos. The accuracy attained was 0.9467 in the experimental scenario. Using the proposed method, the improper use of PPE can be determined without affecting the normal behavior of individuals, which can improve the efficiency of safety management.

1. Introduction

The interaction of humans, machines, materials, methods, environment, and other factors at construction sites complicates the overall management, resulting in frequent accidents. According to the data provided by the Ministry of Housing and Urban-Rural Development (MOHURD) of the People's Republic of China (Ministry of Housing and Urban-Rural Development), the annual number of construction safety accidents in mainland China from 2014 to 2019 was 522, 442, 634, 692, 734, and 773, respectively, with the number of fatalities increasing from 648 in 2014 to 904 in 2019. Falling from height (FFH) and object striking (OS) accidents, the two most frequent types of accidents, accounted for 52% and 14% of the total number of accidents, respectively (Ministry of Housing and Urban-Rural Development). Similarly, according to the Bureau of Labor Statistics (BLS) of the United States, the number of construction fatalities in the US has gradually increased from 933 to 1066 between 2014 and 2019 (Bureau of Labor Statistics), with

approximately 33% of these accidents related to falls (Dong et al., 2021). In addition, a study from 2003 to 2010 showed that fatalities due to traumatic brain injury accounted for 25% of all construction fatalities (Konda et al., 2016).

The United States' National Institute for Occupational Safety and Health (NIOSH) has divided the hierarchy of control (HOC) into five levels. The control effects from the least effective to the most effective are Personal Protective Equipment (PPE), Administrative Controls (AC), Engineering Controls (EC), Substitution, and Elimination (U.S. NIOSH). In the long run, effective control measures can reduce the management costs of construction safety, but the investment costs are also higher in the short term. Elimination, for example, refers to the complete removal of hazards from the system and the environment; it is the most difficult measure to implement, but once achieved, it has the potential to obtain a permanent solution. Since effective control measures are difficult to implement and costly, AC and PPE are the two methods used more frequently, which are also consistent with the management benefits that

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most builders want in short-term construction. Specifically, providing PPE to workers to protect them from physical injury is the easiest control measurement. Several countries have also made PPE mandatory. For example, the General Administration of Quality Supervision, Inspection, and Quarantine (GAQSIQ) in mainland China and the Occupational Safety and Health Administration (OSHA) in the United States have stipulated that a hardhat must be worn during construction work, and a safety harness must be used if the task is performed at heights (GB/T 11651-2008; Occupational Safety & Health Administration). However, even though PPE has been worn, it can still be used and maintained incorrectly, leading to PPE's inability to reduce the number of accident casualties significantly. It can also be seen from the data provided by MOUHUD and BLS that the number of FFH accidents and OS accidents have been at a relatively high level in the past few years, even though the regulations on PPE have been gradually improved over time. Therefore, AC is also needed to regulate workers' behavior and ensure that they are properly using their PPE. External supervision is needed to achieve this measure, which precisely reflects that AC is more difficult to achieve than PPE in the HOC. To address the lack of manual supervision coverage and to control supervision costs, an intelligent inspection method is required to monitor the construction workers' safety and ensure proper use of PPE.

In recent years, a considerable amount of deep learning-related literature in engineering construction has focused on PPE (Du et al., 2011; Dong et al., 2015; Shrestha et al., 2015; Park et al., 2015; Fang et al., 2018; Fang et al., 2018; Fang et al., 2018; Mneymneh et al., 2019; Wu et al., 2019; Wang et al., 2020; Tang et al., 2020; Nath et al., 2020; Chen and Demachi, 2021). However, most studies have only focused on whether PPE is worn (Du et al., 2011; Dong et al., 2015; Shrestha et al., 2015; Park et al., 2015; Q. Fang et al., 2018; W. Fang et al., 2018; Mneymneh et al., 2019; Wu et al., 2019; Wang et al., 2020; Tang et al., 2020; Nath et al., 2020; Chen and Demachi, 2021), overlooking the proper use of the PPE from the perspective of AC in the HOC. In the case of the most frequent FFH and OS accidents, for example, if a worker has not used the hook of the safety harness as required, or has loosened the hardhat, the safety risk would significantly increase. The rapid development of deep learning technology has made it possible to solve this problem. Therefore, we propose a deep learning-based approach for standardized inspection of the use of PPE to promote its proper use from an AC perspective. To this end, we selected two types of unregulated uses of PPE, namely loosening hardhats and not using hooks of safety harnesses before beginning with the task, which may cause significant injuries, specifically when working at heights. We first developed a deep learning-based object detection model for detecting and locating the hardhat and safety harness hooks. The trained computer vision model was used to detect a 9-s video and obtain a document arranged in a time sequence containing the coordinates of the detected hardhat and the safety harness hook. We then extracted the skeleton keypoint data of individuals using another computer vision model, Openpose. Finally, the output results of the first two steps were integrated. A one-dimensional convolutional neural network (1D-CNN) was used to determine whether the video clip contained improper use of PPE.

The remainder of this paper is organized as follows. Section 2 outlines the related works. Section 3 describes the proposed method in detail. Section 4 presents the results based on computer vision models. Section 5 reports the recognition experiments based on the 1D-CNN. Sections 6 and 7 outline the discussion and conclusions, respectively.

2. Literature review

2.1. Deep learning technology application in computer vision

Convolutional neural networks (CNNs) are one of the most common deep learning network structures. Inspired by the biological natural visual cognitive mechanism, CNNs have the characteristics of sharing weights and significantly reducing model complexity. In the traditional

machine learning algorithm, artificial feature extraction is required for input data, whereas CNN can automatically extract data features.

In the 1990s, Lecun proposed the first CNN model, LeNet (Lecun et al., 1998), and this model was used to recognize handwritten numbers in pictures reaching a recognition accuracy of up to 99.05%. Subsequently, the CNN has been continuously improved and optimized, and a variety of forms has been developed such as AlexNet (Krizhevsky et al., 2017), ZFNet (Zeiler and Fergus, 2014), VGGNet (Simonyan and Zisserman, 2014), GoogleNet (Szegedy et al., 2015); and ResNet (He et al., 2016). Initially, CNNs were mainly used in image classification. With the advent of a series of algorithms, CNNs could be applied to object detection. The R-CNN algorithm (Girshick et al., 2014) was of great significance for the use of CNNs for actual object detection. The subsequent Fast R-CNN (Girshick, 2015) and Faster R-CNN (Ren et al., 2017) compensated for the shortcomings of R-CNN and improved the detection speed. The emergence of FCN (Shelhamer et al., 2017) and Mask R-CNN (He et al., 2020) has made image segmentation technology increasingly mature. The advent of lightweight algorithms such as MobileNet-SSD (Howard et al., 2017; Sandler et al., 2018) and You Only Look Once (YOLO) (Redmon et al., 2016; Redmon and Farhadi, 2017; Redmon and Farhadi, 2018; Bochkovskiy et al., 2020; Ultralytics) has made object detection easy on mobile terminals without using cloud servers. Openpose (Cao et al., 2017) is another application of CNNs. Its basic network is VGG-19. Openpose can finally realize the positioning of keypoints and joints of the human body, which can be used for posture estimation and action recognition. The rapid development of deep learning techniques has also led to the rapid advancement of computer vision, with applications in various fields, such as structural health monitoring (Zhang and Yuen, 2021; Adel et al., 2021; Zou et al., 2019; Zhang and Yuen, 2021), medical purposes (Setio et al., 2016; Gulshan et al., 2016), and transportation (Ammour et al., 2017; Huval et al., 2015; Alqizwini et al., 2017).

2.2. Computer vision application in detection of protective equipment

Among the PPE types, hardhats can protect individuals' heads, and they are a primary prevention measure against OS accidents. Safety harnesses can reduce the risk of FFH. This section lists some of the studies that have applied computer vision techniques to identify these two types of protective equipment in recent years.

Du et al. (2011) proposed a method based on machine learning and image processing to identify the hardhat in a video. This method mainly applied Haar-like face features and then detected the hardhat by the color above the face area. Dong et al. (2015) developed a real-time location system and a virtual structure to monitor and evaluate the wearing of PPE automatically. Shrestha et al. (2015) introduced a method for identifying safety hardhats using edge detection algorithms. Park et al. (2015) judged whether the workers in the construction site's video were wearing hardhats based on the spatial position relationship between the detected human body and the hardhat. Q. Fang et al. (2018) and W. Fang et al. (2018) both applied the Faster R-CNN algorithm, with the former applying it to detect whether workers are wearing hardhats and the latter using it to detect the usage of safety harness, and the detection accuracies were 95.7% and 80%, respectively. Q. Fang et al. (2018) also applied a deep learning algorithm to detect the wearing of hardhat and safety harnesses in a high-altitude operation scene that involved working alongside a window and achieved good results in various scenarios. Mneymneh et al. (2019) created a new vision-based safety inspection framework that efficiently and quickly identified construction workers who were not wearing hardhats and issued warnings accordingly. Based on SSD, Wu et al. (2019) proposed a one-stage method based on CNN, which was used to monitor automatically whether construction workers wear hardhats and recognize the corresponding colors with mean average precision (mAP) of up to 83.89%. Wang et al. (2020) presented an end-to-end monitoring network with a MobileNet model for the backbone network, and the method accurately

identified construction personnel who were not wearing hardhats. Tang et al. (2020) proposed a human–object interaction recognition model for PPE verifications, which can identify the usage of a wide range of PPE that is not limited to hardhats and safety harnesses. Nath et al. (2020) introduced three YOLO-based deep learning methods to detect whether construction workers wear safety hardhats and jackets. Chen and Demachi (2021) described a solution to combine object detection and individual detection to identify the improper use of hardhats.

From a HOC perspective, AC is a more effective way to prevent OS and FFH accidents than PPE, and it involves measures such as keeping individuals away from tower crane hooks (Yang et al., 2019), not allowing passage over the support of a construction structure (Fang et al., 2019), prohibiting actions at the edge of a building that is at risk of falling (Yu et al., 2017), and ensuring that protective equipment is used properly. However, although the above-mentioned research results have significantly contributed to this field, none of the studies have focused on the problem of loose hardhats. Regarding safety harnesses, only Fang et al. (2018) noted the possibility of workers wearing a safety harness but not using the hook when working at height. However, that study only identified the hook as a target. In most cases, detecting the hook as a target alone does not indicate that it is being used correctly. In summary, as far as we know, few studies have identified whether safety harnesses and hardhats are being used correctly and in a standardized manner simultaneously from an AC perspective.

3. Methods

Based on the research gap identified from the existing literature, the idea of combining the characteristics of human joints with hardhats and hooks of safety harnesses was formed, and we tried to judge whether PPE was correctly used through the characteristic relationship between them. Therefore, this study proposes a new recognition method that combines an object detection algorithm with the pose estimation algorithm.

First, we collected the research material in the form of videos and defined the behavior of improper use of the hardhat and safety harness hook, which is presented in Section 3.1. Then, an object detection method to recognize and locate the hardhat and safety harness hook is described in Section 3.2. The YOLO v5 algorithm was adopted, which can achieve real-time detection and has good sensitivity to small objects. Section 4.1 describes the dataset for training YOLO v5, which was created by extracting images from the research material introduced in Section 3.1. Section 4.2 is the process of establishing the YOLO v5 model.

The reason for extracting the human joints data is that it can be judged whether the hardhat is loose by the spatial position relationship between the facial features of the human body and the hardhat. It is also possible to evaluate whether the worker has used the hook correctly through the logical relationship between the wrist and the hook's coordinate value. Section 3.3 describes the Openpose algorithm for detecting an individual's skeleton keypoints. Openpose is a classical

algorithm in pose recognition, and it can output up to 25 keypoints with high accuracy and speed. The recognition performance of Openpose was verified in Section 4.3 (Fig. 1).

Finally, when identifying whether a video contains improper use of PPE, it needs to be input to the above-mentioned computer vision algorithms. The coordinates of the hardhat, safety harness, and part of the skeleton keypoints are extracted and integrated. The coordinate values are then converted into relative values and input to the 1D-CNN to complete the final classification. Section 3.4 introduces the relevant principles involved in the 1D-CNN we designed. The reason for selecting 1D-CNN as a classifier is that it can quickly and automatically extract features from the data without manual effort. In Section 5, we describe the process of creating a dataset using the detection data obtained by YOLO v5 and Openpose, and the testing results of 1D-CNN. Fig. 2 is the research framework.

3.1. Collection of videos

This study takes the scene of workers working on the stairs on the top floor of a building, as shown in Fig. 3, as an example to explore the feasibility of the proposed method. In this scenario, the distance between the workers and the falling plane below was 3 m. According to the National Standard of the People's Republic of China GB3608-2008 "Classification of High-Altitude Operations," (GB/T 3608-2008) "Working at heights is defined as the work performed at a high place greater than 2 m above the datum plane for highness of falling, where there is a possibility of people falling down from that high place." It is assumed that the worker has been wearing hardhats and safety harnesses according to the relevant requirements. In this case, before the formal task begins, the worker needs to fix the safety harness hook on a stable end and firmly put on the hardhat. This scenario is selected as a case to detect whether the worker is using these PPE units in a standardized manner.

Table 1 lists the irregularities that may occur concerning hardhats and hooks of safety harnesses. When using the hardhat, the jaw strap is mainly used to fix the hardhat. The main part of the hardhat plays the role of protecting the head. If the main part of the hardhat is loose, it will more likely cause the hardhat to fall off than the loose jaw strap, thus exposing the head to a dangerous environment. Meanwhile, considering that computer vision may be non-sensitive to loose jaw straps, this article selects whether the hardhat is loosened as a case study to determine whether the hardhat is used properly. Currently, there is no clear data standard to define what state of the hardhat is loosened; therefore, the authors define a visible tilt of the hardhat (tilt angle between 30° and 45°) as a loose condition. When the tilt angle is less than approximately 30°, the majority of the head is still in a protected state. As the tilt angle gradually increases, the possibility of the hardhat falling off will also increase. When the tilt angle is greater than about 45°, the protective effect of the hardhat will be significantly weakened. Therefore, we take 30° to 45° as the danger value range to be detected and warned. In the collected video clips, the hardhat's tilt angle is also

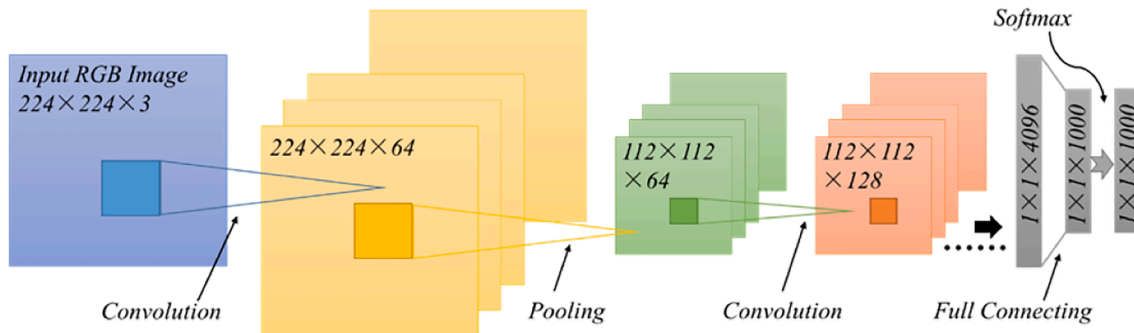


Fig. 1. Structure of convolutional neural network VGGNet.

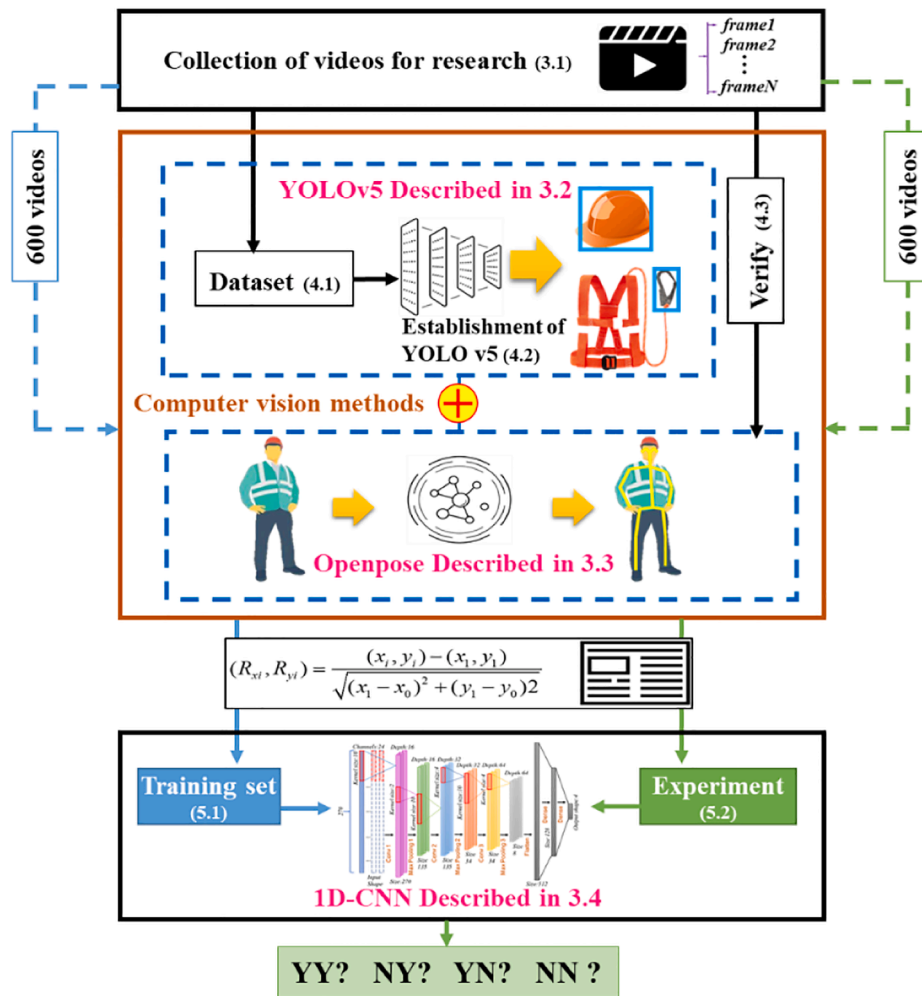


Fig. 2. Framework of this research.

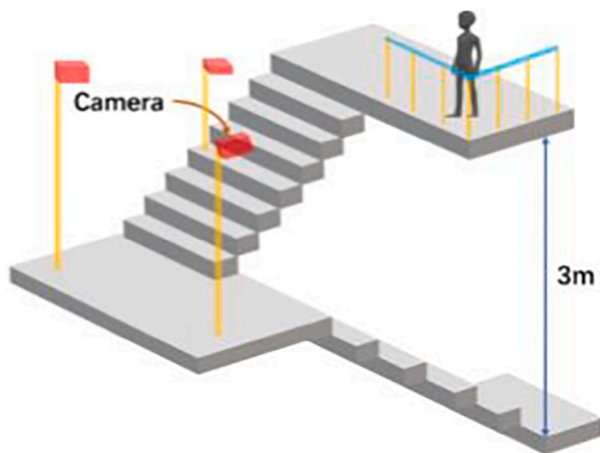


Fig. 3. Simulation of the aerial work scene.

Table 1

Improper behaviors related to hardhat and safety harness.

Hardhat	Loosening the hardhat	Loosening the jaw strap
Safety harness	Not using the hook	Making the straps loose

mainly in this range.

The improper behaviors that may occur when using the safety harness are not using the hook when working to pursue efficiency in the movements and loosening the straps to be more comfortable when moving. Owing to the strap structure, if the strap is only slightly loose, it can still work together with the hook to ensure that workers do not fall. If the strap buckle is not fastened tightly, resulting in a highly loose strap, it increases the risk of falling. However, without the hook, once the worker misses his (or her) footing, the worker will fall resulting in a direct casualty. Meanwhile, considering that computer vision may not be sensitive to loose straps, this study considers whether the safety harness hook is used before the task begins as the research case of whether the safety harness is used properly.

The purpose of this article is to distinguish four situations that may occur before the formal start of the task: 1. The hardhat is properly worn, and the safety harness hook is properly used (YY); 2. The hardhat is not properly worn, but the safety harness hook is properly used (NY); 3. The hardhat is properly worn, but the safety harness hook is not used (YN); 4. The work starts directly without the hardhat and the safety harness hook being properly worn (NN). These abbreviations will be used in the remainder of this article.

In total, 150 videos for each of these four situations were recorded, and then the 600 recorded videos were mirrored to achieve the purpose of data set expansion. Finally, 1200 videos of 9 s in length with a resolution of 1080×1080 and a frame rate of 30 fps were produced, which means that the recorded video contained 30 images per second. Fig. 4 presents an example of the recorded video images for YY and NY. In YY

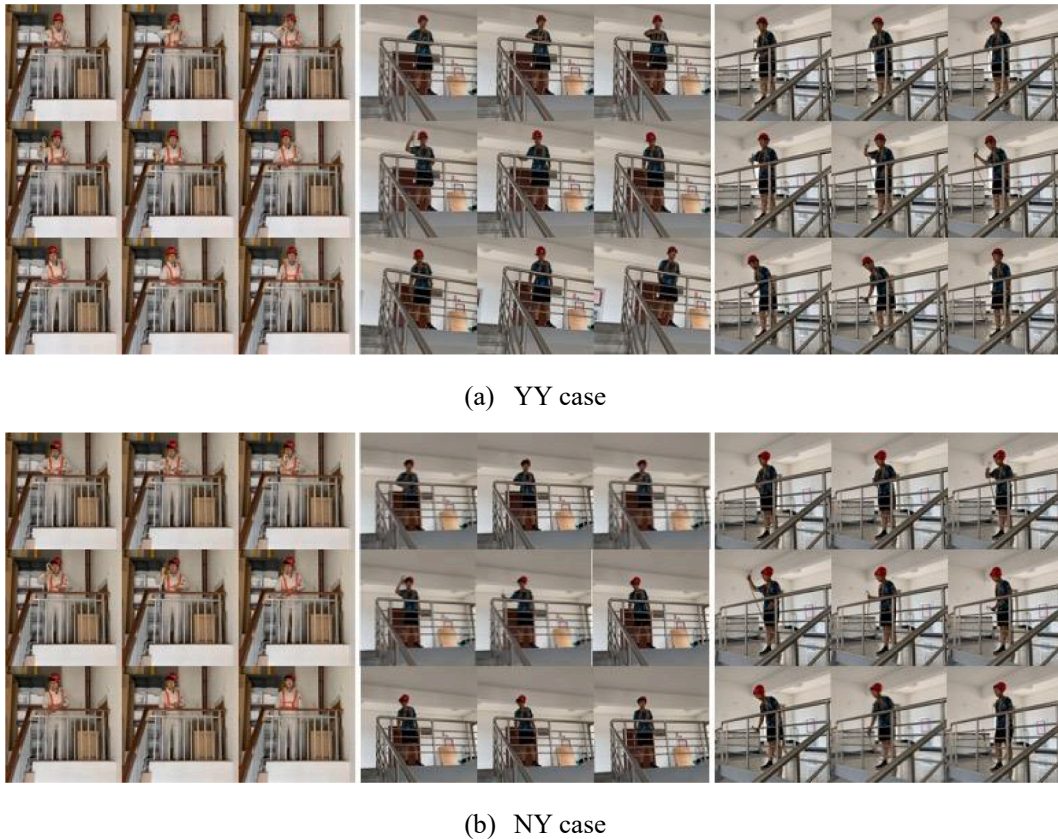


Fig. 4. Conditions of the aerial work scene.

and NN, the character simulated some stair handrail repair or in situ movements without moving the hook to the fixed end.

3.2. Object detection

In general, object detection algorithms can be classified as one- and two-stage. One-stage is represented by the YOLO family (Redmon et al., 2016; Redmon and Farhadi, 2017; Redmon and Farhadi, 2018; Bochkovskiy et al., 2020; Ultralytics), while two-stage algorithms include Faster R-CNN (Ren et al., 2017) and Mask R-CNN (He et al., 2020). The one-stage approach typically uses the backbone network directly to predict the object, without the region proposed network, which is often applied in the two-stage approach and therefore is faster. In addition, after continuous development in recent years, the accuracy of one-stage approaches has been enhanced significantly. Therefore, in this study, YOLO v5, the most advanced one-stage algorithm, was selected as the object detection algorithm to recognize the hardhats and safety harness hooks.

YOLO v1 was proposed by Redmon in 2016 (Redmon et al., 2016). It divided the image into 7×7 grids and then predicted the objects in the grid directly by a CNN. However, since each grid can only predict one object, the overlapping objects are challenging to detect, and small objects are likely to be undetected. The subsequent YOLO v2 (Redmon and Farhadi, 2017) introduced batch normalization, which normalizes the input of each layer and makes convergence easier. In terms of network structure, YOLO v2 applied DarkNet-19 and eliminated the fully connected layer. YOLO v2 also introduced anchor boxes, making it possible to predict more objects. Nevertheless, as YOLO v2 only considered the output after the fifth down-sampling, it failed to solve the problem of difficult detection of small objects.

The most significant improvement of YOLO v3 (Redmon and Farhadi, 2018) was the modified network structure, which applied DarkNet-53 to achieve down-sampling by convolution with a stride of

two instead of pooling layers. When outputting the results, large, medium, and small objects were predicted using three scales. When designing anchor boxes, different sizes were also designed for large, medium, and small objects to improve the detection performance of small objects. Meanwhile, for better feature extraction, YOLO v3 introduced a residual connection.

To further enhance the performance of YOLO v3, YOLO v4 (Bochkovskiy et al., 2020) and v5 (Ultralytics) have made several improvements. They have in common the use of Mosaic data enhancement, use of the CSPDarkNet-53 as the backbone, use of a feature pyramid network (FPN) and path aggregation network (PAN) structure in Neck, and application of CIOU as the loss function. The difference is that YOLO v5 uses adaptive anchor box computation at the input level, applies the focus structure in backbone, introduces cross-stage partial connections (CSP) in Neck, and uses SiLU as the activation function.

Fig. 5 shows a flow chart of YOLO v5. When predicting the image, the original image is first scaled to 640×640 , and after the features are extracted by CSPDarkNet and spatial pyramid pooling, it is processed by FPN and PAN. This method makes the high-level and low-level features better fused, thus improving the accuracy. Finally, objects are output from three scales to recognize large, medium, and small objects. In this study, we use the weights of YOLO v5m in training the detection model of PPE, and the m and n in Fig. 5 are taken as 0.75 and 0.67, respectively.

3.3. Openpose

Openpose is one of the state-of-the-art algorithms in pose estimation, which can output up to 25 keypoints of an individual (Cao et al., 2017). It also has the function of finger movement estimation and facial expression estimation. We selected it as the output tool for skeleton keypoints owing to its advantages of fast processing speed and high accuracy.

Openpose is a bottom-up human posture recognition method that

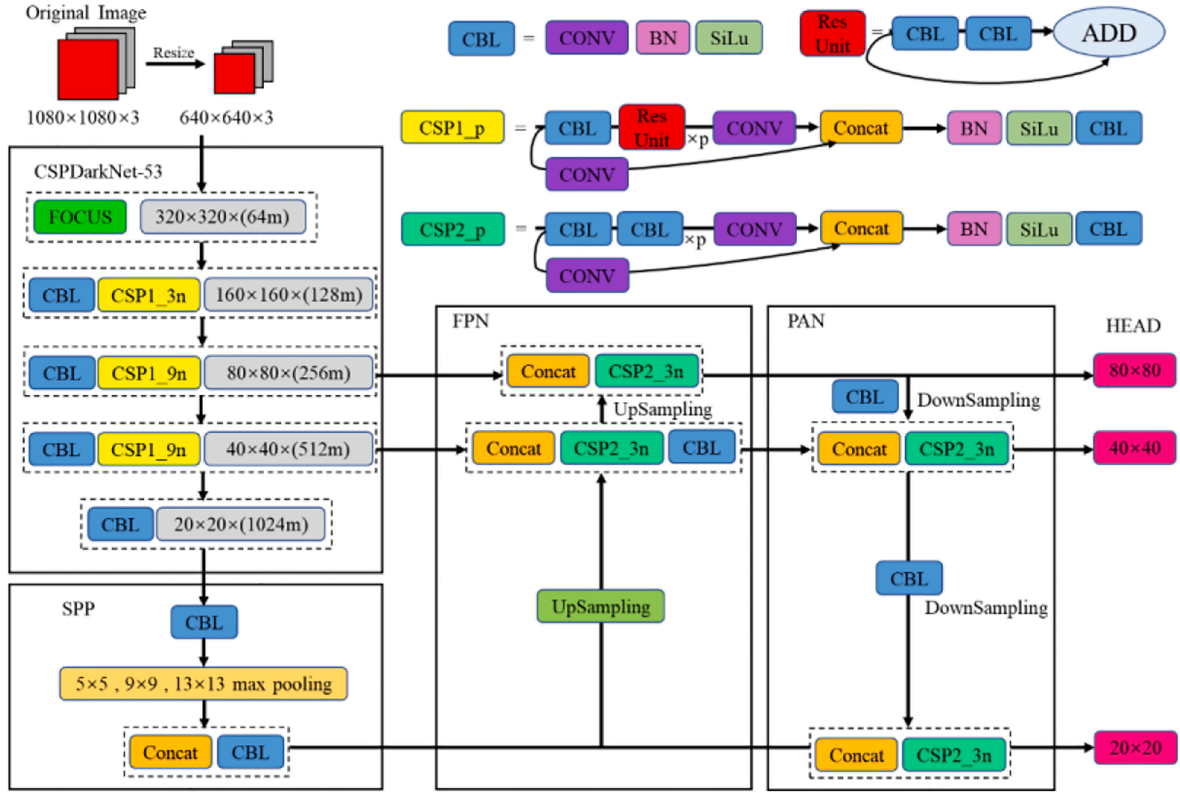


Fig. 5. Network structure of YOLO v5.

can realize pose estimation of human motion, facial expression, and finger movement (Cao et al., 2017). Openpose first uses the first ten layers of the VGG-19 CNN to extract the F feature from the input image. Then, feature F is processed through a continuous multistage network. Each stage (t) of the network contains two branches. The output results are the part confidence map (S^t) and part affinity map (L^t). Finally, the feature representation of the part affinity fields (PAFs) is used to save the position and direction information of the body's support area. As shown in Fig. 6, Openpose uses an iterative CNN network to detect key parts of the human body. Each CNN network has two branches, namely CNN_S and CNN_L. The networks in the first and subsequent stages differ in form. The two network branches at each stage are used to calculate the part confidence maps and part affinity maps. It can also be observed from the figure that the input feature F received in the first stage of the network is processed by the network to obtain S^1 and L^1 . From the second stage, the input of the stage (t) network includes three parts: S^{t-1} , L^{t-1} , and F . The input of each stage network is given by Eq. (1) (Cao et al., 2017):

$$\begin{aligned} S^t &= \rho^t(F, S^{t-1}, L^{t-1}), \forall t \geq 2 \\ L^t &= \phi^t(F, S^{t-1}, L^{t-1}), \forall t \geq 2 \end{aligned} \quad (1)$$

In the procedure, iterate repeatedly until the network converges. Next, determine whether the two parts (d_{j1} and d_{j2}) are connected through PAFs and calculate the linear integral on the line of d_{j1} and d_{j2} . If the direction of $\vec{d_{j1}d_{j2}}$ is consistent with that of $L^*_c(p)$, the value of the linear integral E (Cao et al., 2017) will be very large, and these two parts are likely the torso.

$$E = \int_{u=0}^{u=1} L_c(P(u)) \cdot \frac{d_{j2} - d_{j1}}{\|d_{j2} - d_{j1}\|_2} du \quad (2)$$

Finally, traverse all the collocations, calculate the integral sum, and determine the torso of each part of the body. Two adjacent torsos must have shared joint points, and all the skeletons can be obtained by combining all the torsos through the joint points. Fig. 7 shows the image of the joint points obtained after applying Openpose, illustrating the position represented by each keypoint number.

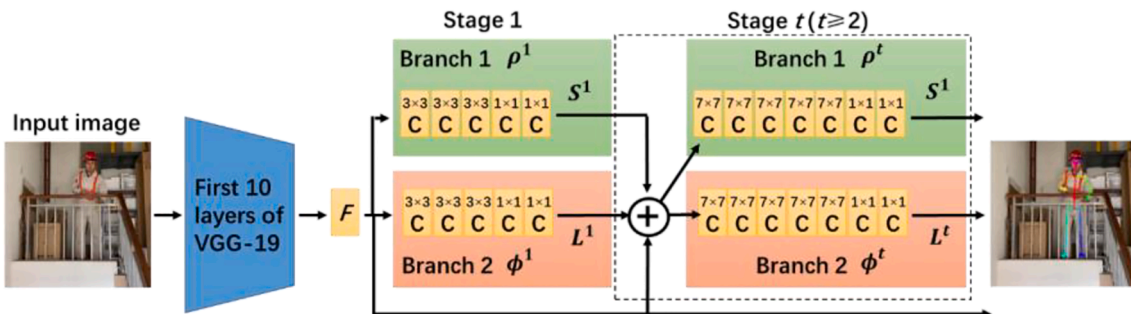


Fig. 6. Network architecture of keypoints detected by Openpose.



Fig. 7. Individual skeleton keypoints detected by Openpose.

3.4. One dimensional convolutional neural network

Each video recorded in this article contained 270 images. Input a video into the YOLO v5 and Openpose model, and then a 270×8 and 270×75 data files will be obtained, respectively. The 8-column data of the former are the coordinates of the hardhat and safety harness hook, which are Y_{max} , Y_{min} , X_{max} , and X_{min} . The 75-column data of the latter are the X coordinates, Y coordinates, and confidence scores of the 25 keypoints.

Fig. 8 shows a sample of the worn properly and worn improperly hardhats in this study. In Section 3.1, we have already mentioned that when there is a significant tilt of the hardhat that may cause falling off (tilt angle between approximately 30° and 45°), it can be considered a case of loose hardhat. It can be observed that when a hardhat is loose, its coordinates in the image will change, and the relative distance between it and the individual's skeleton keypoints will also change. Among the 25 keypoints output by Openpose, the distance of the hardhats from the human eye and ear is closer than that from other parts, and thus the variation in the relative distance between the hardhat and the eye as well as the ear will be more significant. Therefore, the coordinates of the eyes and ears (keypoints 15 to 18) are extracted to enable the 1D-CNN to extract the change features of the data better when the hardhat is loose.

If the worker uses the hook correctly, the coordinates around the wrist (keypoints 3, 4, 6, and 7) and the hook will reflect the same trajectory. If the hook is not used, it will not be reflected. Finally, the coordinates of points 3, 4, 6, 7, 15, 16, 17, and 18 from the data output by

Openpose are extracted and then summarized with the results output by YOLO v5.

In this study, there were 1200 270×24 data files. If traditional machine learning algorithms are used to identify these data, manual feature extraction is required, which is cumbersome. The convolutional and pooling layers of the CNN can directly extract features, eliminating the need for manual feature extraction. Although traditional CNNs are generally used to deal with RGB image issues, images are essentially matrix problems. Therefore, it is feasible to apply CNNs to identify arrays of these data (Harbola and Coors, 2019; Lu et al., 2020; Xu et al., 2020).

Table 2 and Fig. 9 present the network structure of the 1D-CNN designed in this study. The input size is 270, with a total of 24 channels through three convolutional layers and three pooling layers, and then through the flattened layer and two dense layers, to achieve the final output classification. Unlike the traditional CNN used to process image issues, the kernel size of the convolutional layer and pooling layer of the CNN designed in this study has only one dimension. Therefore, it is called a 1D-CNN.

3.4.1. Convolution layers

The purpose of setting a convolution layer is to expand the longitudinal depth of the matrix. The convolution layer consists of a convolution kernel. The depth of the kernel must be the same as that of the input matrix. Taking the Conv2 layer as an example, the input size is 135, depth is 16, size and depth of the convolution kernel are 10 and 16, respectively, and kernel number is 32. Once there is convolution, the kernel moves one step from top to bottom and convolutes with the input matrix to generate a new matrix with a depth of 32. If the "Padding" is the same, the output vector's length is the same as the input. If the "Padding" is Valid, the length of the output is $135 - 10 + 1 = 126$. Fig. 10(a) illustrates the principle of the convolution layers with a simple column vector.

3.4.2. Pooling layers

The main function of the pooling layer is to reduce the dimensions of the input matrix and extract features. As mentioned before, when the Conv2 operation is completed, the matrix is input into the pooling layer. The size is 135, and the kernel size and stride are all four. From top to bottom, a maximum value is selected for every four elements. If the "Padding" is the same, 135 is filled to 136, which can be divided by 4, and the output size is 34. If the "Padding" is valid, the integer part after division is taken, that is 31. Fig. 10(b) explains the pooling layer principle with the output shown in Fig. 10(a).

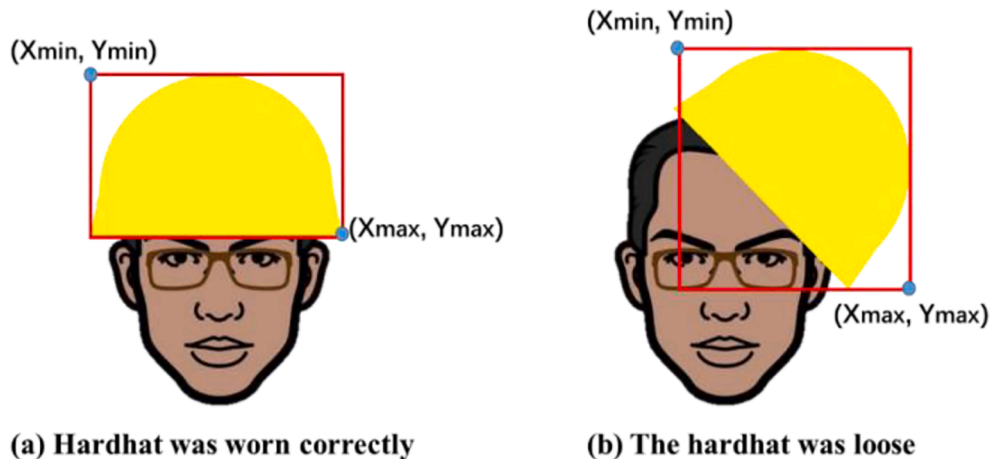
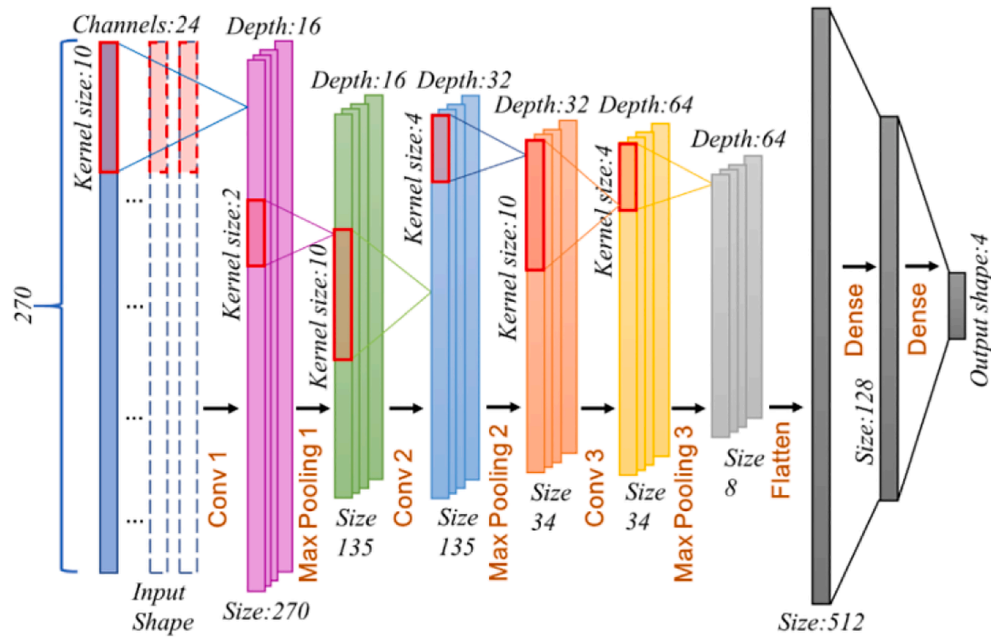
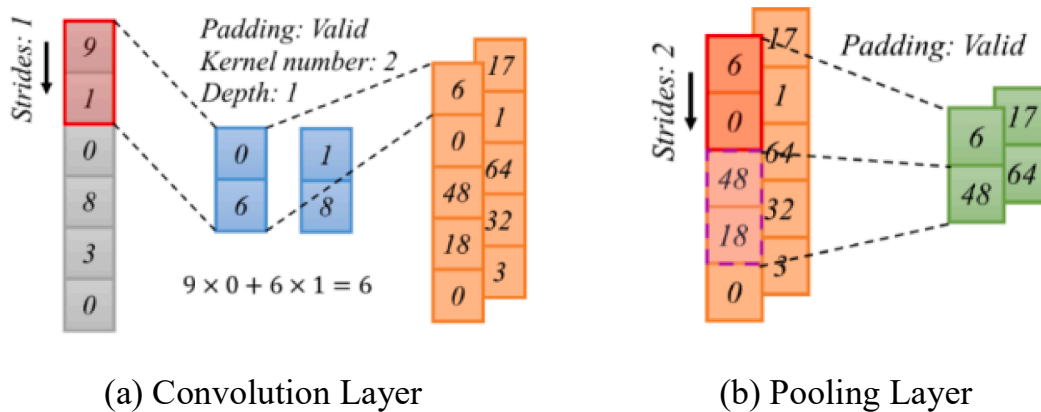


Fig. 8. Relative position relationship between the hardhat and eyes.

Table 2

Details of the designed one-dimensional convolutional neural network (1D-CNN).

Layers	Input shape	Kernel size	Kernel number	Strides	Padding	Activation	Output shape
Conv1	(270,24)	10	16	1	Same	Relu	(270,16)
Pooling1	(270,16)	2	None	2	Valid	None	(135,16)
Conv2	(135,16)	10	32	1	Same	Relu	(135,32)
Pooling2	(135,32)	4	None	4	Same	None	(34,32)
Conv3	(34,32)	10	64	1	Same	Relu	(34,64)
Pooling3	(34,64)	4	None	4	Valid	None	(8,64)
Flatten	(8,64)	None	None	None	None	None	(512)
Dense	(512)	None	None	None	None	Relu	(128)
Dense	(128)	None	None	None	None	Softmax	(4)

**Fig. 9.** Structure of the 1D-CNN.**Fig. 10.** Principles of convolution layer and pooling layer.

3.4.3. Dense layer

The dense layer plays the role of “classifier” in the entire CNN. Suppose the operations of the convolution layer, pooling layer, and activation function are used to map the original data to the hidden layer feature space. In that case, the dense layer plays the role of mapping the learned “distributed feature representation” to the sample label space. In Fig. 9, after Pooling3 and Flatten, the input data are expanded into a vector of 1×512 , and Dense1 processes 512 nodes to generate a vector

of 1×128 . Then, the final vector of size four is obtained by the Softmax function in Dense2, corresponding to YY, NY, YN, and NN.

3.4.4. Activation function

The activation function exists in the convolution and dense layers. The activation function activates a portion of the neurons in the neuron network and transmits the activation information back to the next layer of the neural network. It is used to add nonlinear factors to compensate

for the inadequate expression of the linear model. As shown in Fig. 11, the common nonlinear activation functions are sigmoid, tanh, and ReLU. Currently, ReLU is one of the most popular functions.

4. Results based on computer vision algorithms

4.1. Dataset for YOLO v5

The dataset used in this study to train the YOLO v5 model contained 3208 images. Of these, 50% were from the originally recorded videos. The other 50% were obtained from the videos after horizontal mirroring to expand the dataset and avoid overfitting. As presented in Table 3, the videos of YY and NY contain various statuses of safety harness hooks and hardhats. Therefore, the images of the dataset were all selected from YY and NY, of which there were 1604 images in each. Table 4 lists the source of the dataset images.

Since the video recording uses a frame rate of 30 fps, once the hook of the safety harness exceeds this rate when moving with the arm, the moving object will be blurred in the video. To ensure that the trained YOLO v5 has better detection capabilities for blurred moving objects, a total of 2182 images of the safety harness hook in various positions in the air were considered in the dataset. In addition, because of the application of Mosaic data augmentation, the possibility of overfitting in YOLO v5 is further reduced.

4.2. Detection results of YOLO v5 model

For training and validation, 80% and 20% of the images in the dataset were used, respectively. The parameters used to train YOLO v5 were as follows: initial learning rate of 0.01, batch size of 8, and 500 epochs. The average precision (AP) and mAP were used to evaluate object detection performance. The numerical values of AP and mAP are closely related to precision and recall. The calculation method for precision and recall is shown in Eq. (3) (Powers, 2011):

$$\begin{aligned} \text{precision} &= \frac{TP}{TP + FP} \\ \text{recall} &= \frac{TP}{TP + FN} \end{aligned} \quad (3)$$

According to the above equations, when a validation set image contains a hook, if the detection model correctly detects it as a hook, the image is marked as a true positive (TP). If the detector mistakenly judges the object as a hardhat, the image is marked as a false negative (FN). If the detector misjudges the hardhat as the hook, the image is marked as a false positive (FP) for the hook.

When the intersection over union (IOU) is greater than a certain value, the result is output. When the IOU threshold is set differently, the precision and recall values will also change; therefore, the two parameters cannot be used to evaluate the performance directly. After training,

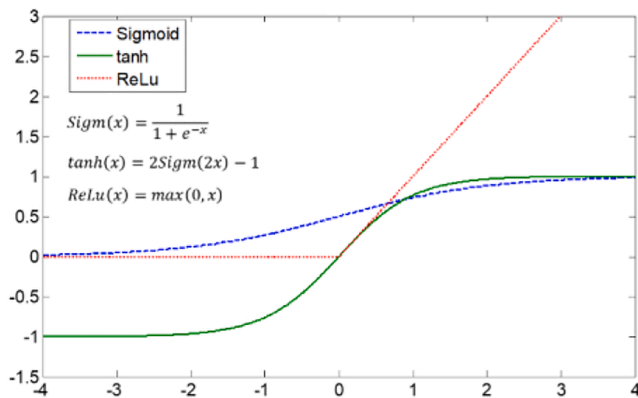


Fig. 11. Common activation functions.

Table 3
PPE status.

	YY	NY	YN	NN
Harness's hook	Original position, in movement, fixed	Original position, in movement, fixed	Original position	Original position
Hardhat	Firm	Loose	Firm	Loose

Table 4
Source of images in dataset.

	Hook position of YY		Hook position of NY	
Number of images	Original Position	250	Original Position	260
	In movement	1102	In movement	1080
	Fixed	252	Fixed	264

all the images of the validation set were tested. The precision and recall were calculated for different IOUs, and their relationships were determined. When the precision-recall curve envelope area is close to 1, the trained model has better detection and classification performance; this area is also called AP. From the AP, the mAP can be obtained, which represents the overall detection ability of the model. In this study, the threshold IOU was set to 0.5 to obtain AP@50 and mAP@50.

To validate the effectiveness of the YOLO v5 model, two classical object detection algorithms, SSD (Sandler et al., 2018) and Faster R-CNN (Ren et al., 2017), were selected for comparison in this study. The AP, mAP, and time consumed to detect a single image are listed in Table 5. It can be observed that YOLO v5 has an advantage in both detection accuracy and speed. Fig. 12 presents the detection results for the images corresponding to Fig. 4. In most cases, the trained detection model can accurately locate and detect the hardhat and safety harness hook from different angles. Although there is still unrecognition when the hook moves too fast, the beginning and end positions can be correctly positioned, and the conditions for subsequent research are available.

Then, a video from each of the four types was selected and input into the trained detection model. Fig. 13 shows the relationship between the coordinates of the detected boxes with time. It can be observed that the coordinates of the hook in (a) and (b) are significantly different from those in (c) and (d). This is because the hook will have a moving track when workers use it correctly. It can also be found that the loose hardhat does not show apparent characteristics in these curves.

4.3. Recognition results based on Openpose

Although the difference between the hook coordinates can be seen in Fig. 13, its trajectory does not fully explain its correct use. It needs to be combined with the upper limb coordinates in the skeleton keypoints for further confirmation. Moreover, when the hardhat is tilted and loosened, no apparent features can be observed from the curves in Fig. 13 alone. As mentioned earlier, when the hardhat is loose, the relative positional relationship between it and the five sense organs will change significantly; therefore, we must extract the five sense coordinates contained in Openpose. Therefore, in this section, we use the open-source algorithm Openpose described in Section 3.3, and input the video into it to obtain images with keypoints. The output results of the images corresponding to Fig. 4 by Openpose are listed in Fig. 14, and it can be observed that each predicted keypoint is in good agreement with

Table 5
Average precision (AP) and mean average precision (mAP).

	AP-hardhat	AP-hook	mAP	time
YOLO v5	0.996	0.986	0.991	0.019 s
SSD	0.981	0.943	0.962	0.037 s
Faster R-CNN	0.999	0.975	0.987	0.166 s

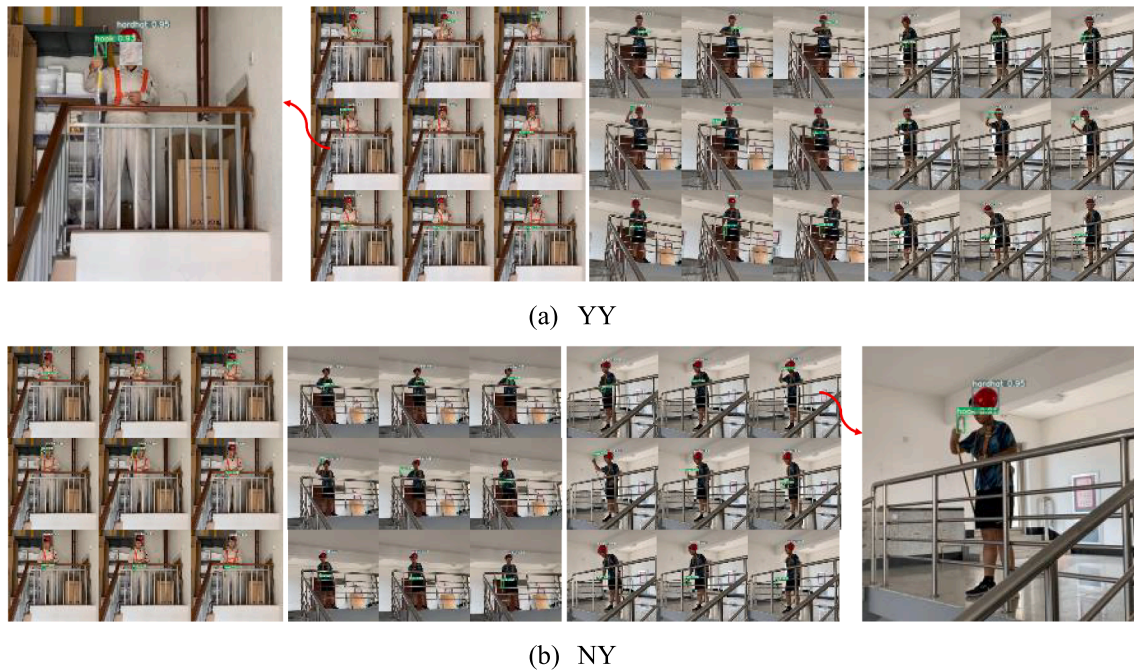


Fig. 12. Detection results.

the actual situation.

Fig. 15 shows the coordinate curves of keypoints 3, 4, 6, 7, 15, 16, 17, and 18 after the four videos corresponding to Fig. 13 are output by the Openpose algorithm. The figure shows that when the safety harness hook is used correctly, the coordinate values of RWrist_Y and Hook_Y present a similar shape. Moreover, the coordinates of the eyes and ears also present a shape similar to the coordinates of the hardhat above.

When Openpose outputs the coordinate value, it also outputs the confidence score of this frame. The closer the score is to 1, the closer it is to the true value. Each video will have 270 confidence scores. For each video, the average confidence score of these 8 points is calculated. Then, based on 300 videos of each category, the mean confidence score of each point is calculated. These mean values are listed in Table 6. It can be observed that the confidence score of each point is at a relatively high level. Combining the results listed in Figs. 14 and 15, it can be considered that the coordinate values output by Openpose are reliable and can be applied to the next step.

5. Experiment based on one-dimensional convolutional neural network

5.1. Dataset

The curves in Figs. 13 and 15 show only a few apparent features, most of which are difficult to understand, especially on the issue of whether the hardhat is loose. Therefore, the 1D-CNN is designed with a strong ability to extract data features as a tool for subsequent analysis.

As mentioned above, 300 video clips for each category were recorded for a total of 1200 videos. From these, 600 videos were selected to train the 1D-CNN in Fig. 9 and the other 600 videos to test the classification performance. Before making the dataset, the trained YOLO v5 and Openpose models were used to detect these videos and output coordinate data. Fig. 16 shows the detection results for the same video using these two algorithms.

After inputting all the videos, the X and Y coordinates of points 3, 4, 6, 7, 15, 16, 17, and 18 in Fig. 16(a) and all coordinate values in Fig. 16 (b) are extracted and integrated. As shown in Fig. 17, when the worker's location in the video is different, and the distance between the worker and the lens is different, the output coordinates will be quite different.

Therefore, relative values must be adopted when creating the datasets. The calculation method for the relative values is shown in Eq. (4). To eliminate the influence of the worker's position, we subtracted the X and Y coordinates of point 1 output by Openpose from a total of 24 sets of extracted coordinate values. Subsequently, to eliminate the influence of the distance between the worker and the lens, all relative coordinate values were divided by the distance between point 0 and point 1 output by Openpose. Finally, as shown in Fig. 18, 24 training set files corresponding to 24 channels were obtained, and each training set file was an array of 600×270 . The first number, 600, is composed of 150 sets of data contributed by each category, and 270 is the coordinate value of the 9 s video output.

5.2. Experiment results

Of the data in the training set, 90% were used to train and fit the model. When the model was established, the remaining 10% was used to evaluate the trained model preliminarily. The batch size represents the number of samples trained in each step of the CNN. In this study, the batch size was 4. The learning rate determines the speed at which the model converges. In this study, the learning rate was set at 0.0001, and the number of epochs was set at 300. One epoch was equal to one training session with all the samples in the training set. There were 540 sets of data used for training, and the number of iterations was 40000. Figs. 19 and 20 show the loss and accuracy curves versus the iterations, respectively. Fig. 19 shows that as the number of training iterations increases, the loss value decreases, stabilizing at the end of the training. It can be observed that the network model converged. As depicted in Fig. 20, the accuracy continues to increase with iteration. By the end of the training, the training accuracy and validation accuracy reached 1.00 and 0.95, respectively.

Finally, another 600 videos that were not involved in the training set were used to test the trained 1D-CNN model. The test results are presented in Tables 7–10. From the data in the table, the trained 1D-CNN model shows good recognition ability. The precision of each category and the recall values do not appear to be significantly different, and both reach 0.9 or more. Most of the errors that occur are reflected in the detection of whether the hardhat is worn loose. The final recognition accuracy rate is 0.9467. The recognition accuracy of the standardized

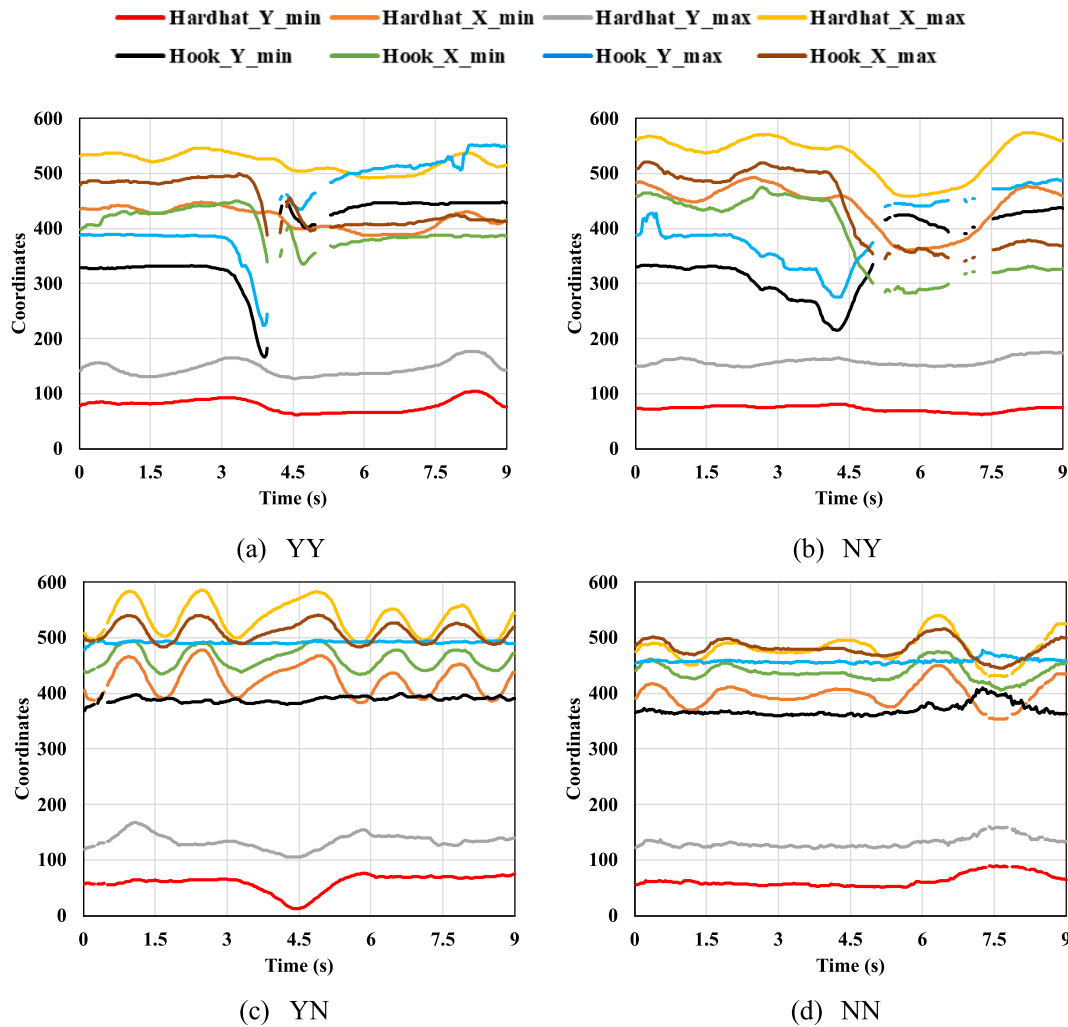


Fig. 13. Curves of coordinates of hardhat and hook versus time.

use of hardhats and safety harnesses reached 0.9531 and 0.9965, respectively. Considering that the ratio of the training set to the test set in this study reached 1:1, which is different from the 8:2 or even higher set by most researchers, this also illustrates the reliability of the trained model and the proposed method.

6. Discussion

The main contributions of this study are as follows:

In terms of safety management, previous studies on PPE are mostly limited to the lowest level of the HOC, that is, detecting whether workers are using PPE, and the higher levels are not applied to monitor if the PPE is being used correctly. In the long term, the lower the level in the HOC, the higher is the management cost. Although advanced computer vision technology has eased the management pressure to a certain extent, the number of casualties every year due to FFH and OS accidents is still high. Hence, it is necessary to focus on whether PPE is being used properly at the AC level. The method proposed herein addresses the problem of PPE monitoring from the AC perspective. It can detect the loosening of hardhats on the head and the non-use of harness hooks when working at height, two types of improper use of PPE that can easily result in injuries to the workers themselves. In practical application, it can be combined with safety training to require workers to be checked for standardized use of PPE by the method we propose. When a worker begins the task directly without using PPE properly, he or she can be alerted and warned.

In terms of the theoretical approach, we realize that it may be challenging to obtain good results for the problems presented in this paper if only object detection algorithms are used. For example, in some cases, from a two-dimensional image, if the hook of the safety harness at the fixed end happens to be located in front of the chest, it will likely be confused with the case where the safety harness hook is not used. Noting the relationship between the positions of the hardhat and the head and the hook of the safety harness and the hand, a framework for fusing the object detection algorithm with the individual's pose estimation algorithm is proposed to further improve the theoretical approach related to PPE.

On the technical side, the object detection algorithm is used for the detection of two categories only, the hardhat and safety harness hook, and the skeleton keypoint algorithm is used to determine the location of the keypoints related to the PPE. A 1D-CNN is designed to classify the obtained data, which can detect the improper use of PPE in video clips from three angles under the conditions pre-defined in this study. The accuracy reached 0.9531 for the loosening of the hardhat on the head and 0.9965 for the issue of not using the hook of the safety harness. For comparison, we selected two classical classification algorithms, BP neural network (Rumelhart et al., 1986) and support vector machine (SVM) (Cortes and Vapnik, 1995). The input data of the BP neural network and SVM are the feature values that need to be extracted manually. In this study, 17 feature values, including mean, variance, kurtosis, skewness, and others, were extracted for the classification. Table 11 lists the results. It can be observed that the classification



(a) YY



(b) NY

Fig. 14. Output by Openpose.

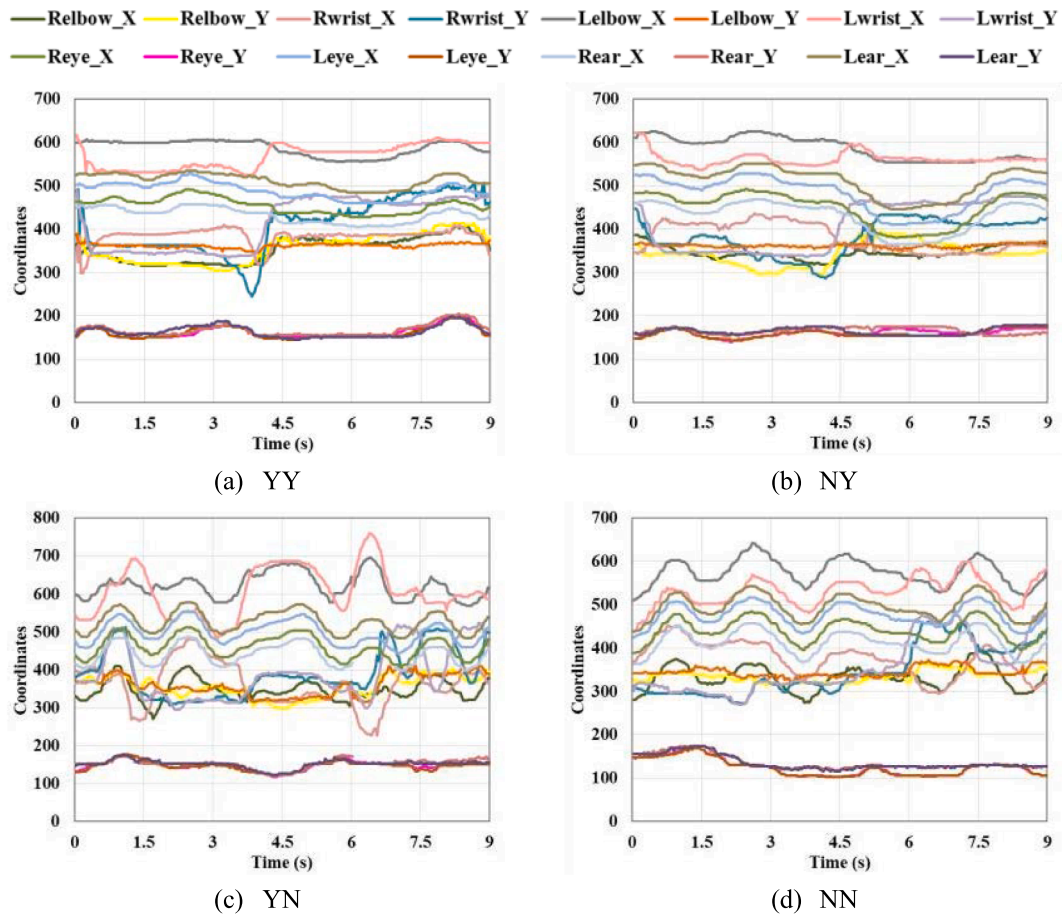
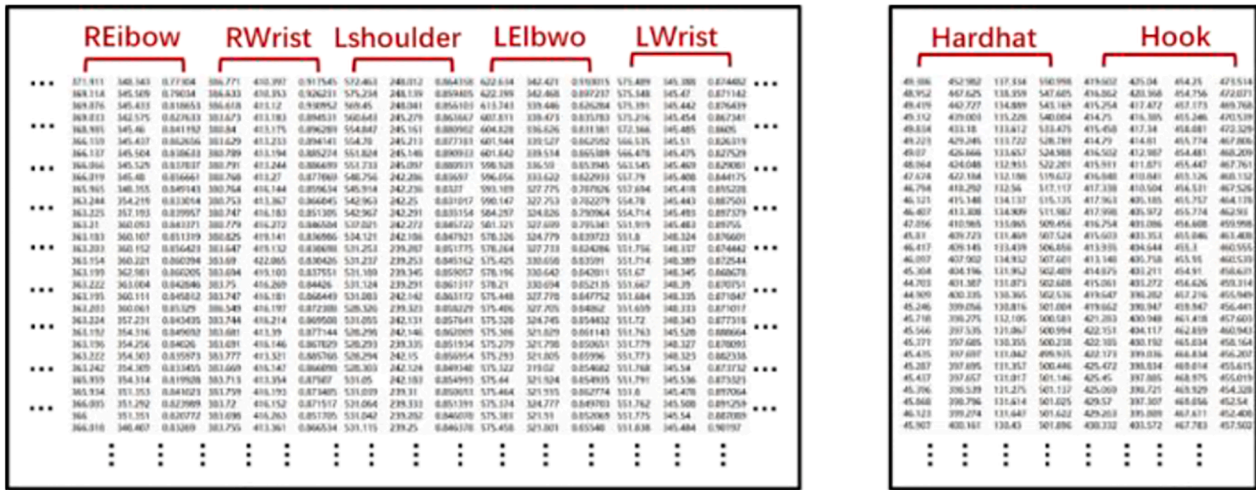


Fig. 15. Curves of coordinates output by Openpose versus time.

Table 6

Mean average confidence score of each keypoint.

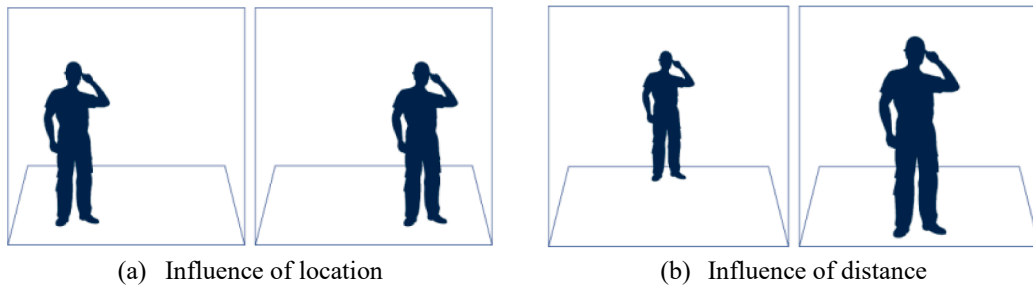
	Relbow(3)	Rwrlist(4)	Lelbow(6)	Lwrist(7)	Reye(15)	Leye(16)	Rear(17)	Lear(18)
YY	0.8100	0.8077	0.8112	0.8056	0.8937	0.9008	0.7959	0.8134
NY	0.8155	0.8193	0.8150	0.8149	0.9029	0.9058	0.7814	0.7937
YN	0.7996	0.7891	0.7738	0.7655	0.8901	0.8915	0.7708	0.7738
NN	0.8368	0.8394	0.8300	0.8340	0.9064	0.9089	0.7958	0.7952



(a) Coordinates output by Openpose

(b) Coordinates output by YOLO v5

Fig. 16. Coordinates output by Openpose and YOLO v5.



(a) Influence of location

(b) Influence of distance

Fig. 17. Influence factor.



Fig. 18. Details of the training set.

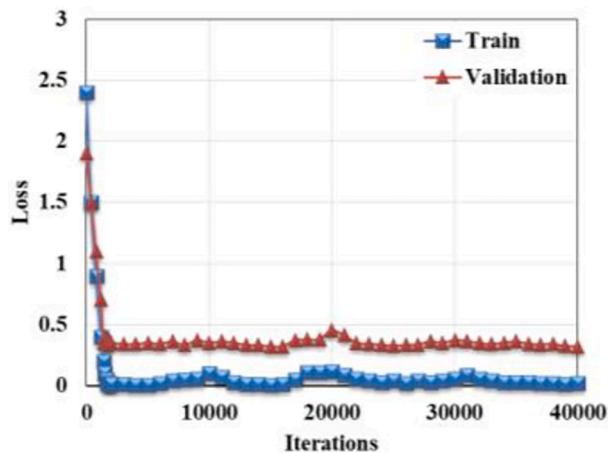


Fig. 19. Loss curves.

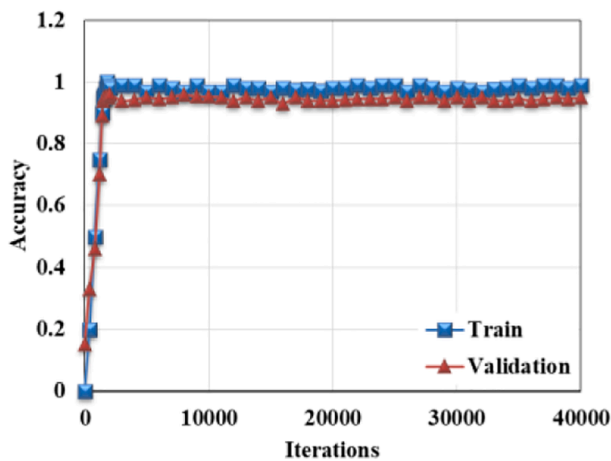


Fig. 20. Accuracy curves.

Table 7
Test results of the 1D-CNN.

Recognized as	YY	NY	YN	NN
YY	143	7	1	0
NY	7	142	1	1
YN	0	1	140	6
NN	0	0	8	143
Accuracy	0.9467			

Table 8
Precision and recall value.

	YY	NY	YN	NN	Average
Precision	0.9533	0.9467	0.9333	0.9533	0.9415
Recall	0.947	0.9404	0.9523	0.947	0.9467

Table 9
Test results for standardized use of hardhat.

Recognized as	YY	NY	Recognized as	YN	NN
YY	143	7	YN	140	6
NY	7	142	NN	8	143
Accuracy	0.9532		Accuracy	0.9529	
Mean Accuracy	0.9531				

Table 10
Test results for standardized use of safety harness.

Recognized as	YY	YN	Recognized as	NY	NN
YY	143	1	NY	142	1
YN	0	140	NN	0	143
Accuracy	0.9965		Accuracy	0.9965	
Mean Accuracy	0.9965				

Table 11
Comparison results for data classification.

	Accuracy	Accuracy-hardhat	Accuracy-safety harness
1D-CNN (ours)	0.9467	0.9531	0.9965
BP Neural Network	0.775	0.8033	0.9710
SVM	0.703	0.707	0.97

accuracy of 1D-CNN is significantly higher, and 1D-CNN can automatically extract features during prediction, which is not available in machine learning algorithms. It can be seen, the results obtained by applying the method designed in this study validate the feasibility of using the coordinates of protective equipment and skeleton keypoints to identify the standardized use of PPE, further illustrating the effectiveness of using 1D-CNN to deal with the data classification problem.

Tables 12 and 13 present a comparison between some of the deep learning-based studies related to PPE and this paper. As can be observed, the purpose and methodology of this study are different from those of previous studies. Our method can identify whether PPE is used properly from the perspective of AC, and the results obtained are comparable.

However, this study also has some limitations and shortcomings that need to be addressed in the future.

The first is the dataset. A complete dataset will help to apply the results of this study to actual projects more quickly. The current research object of this study uses stair work as an example. In the future, multi-person work and different scenes should be considered in the dataset, and more different angles and different light conditions should be covered.

Second, in the future, to solve the problem that the hook is prone to target loss when the moving speed is too fast, we can try to use the object tracking method or use a camera with a higher frame rate.

Finally, in terms of application objects, the problem of irregular use not only affects hardhats and safety harnesses but also goggles, protective clothing, gloves, etc. Therefore, in the future, it is necessary to extend the method developed in this study to the inspection of various types of PPE to improve the safety of the construction site comprehensively. In addition, when using hardhats and safety harnesses, it also has the problem of loose jaw straps and shoulder straps, which are not considered in this paper because the computer vision technology is not

Table 12
Different deep learning-based studies related to hardhat's monitoring.

	Objective	Method	HOC	Accuracy
Fang et al. (Fang et al., 2018)	Identify whether the worker is wearing a hardhat;	Faster R-CNN	PPE	0.957
Wang et al. (Wang et al., 2020)	Identify whether the worker is wearing a hardhat;	MobileNet with with Multi-Scale Features and a Top-Down Module	PPE	0.894
Nath et al. (Nath et al., 2020)	Identify whether the worker is wearing a hardhat;	YOLO v3	PPE	0.7397
Ours	Determine if the hardhat is loose when the hardhat has been worn	YOLO v5 + Openpose + 1D CNN	AC	0.9531

Table 13
Different deep learning-based studies related to safety harness's monitoring.

	Objective	Method	HOC	Accuracy
Fang et al. (Fang et al., 2018)	Identify whether the worker is wearing a safety harness;	Faster R-CNN + Harness detection network	PPE	0.8
Chen et al. (Chen and Demachi, 2021)	Identify whether the worker is wearing a safety harness;	YOLO v3 + Openpose + Scene graph identification	PPE	0.9337
Ours	Determine if the hook is used when the safety harness has been worn	YOLO v5 + Openpose + 1D CNN	AC	0.9965

sensitive enough to them. Therefore, it is necessary to explore a feasible and effective method to address this issue in the future.

7. Conclusion

PPE is considered to be the least effective control measure in the hierarchy of controls. To improve the effectiveness of PPE and to reduce the frequency of FFH and OS accidents, it is necessary to monitor the behavior of using PPE from the perspective of AC. The authors took the FFH and OS accident as the research background and selected a high-altitude staircase working scene as a research case, proposed a computer vision-based standardized use inspection method of PPE.

First, four usage scenarios of PPE are defined, including one standard behavior and three non-standard behaviors, namely: 1. The hardhat is properly worn, and the safety harness hook is properly used (YY); 2. The hardhat is not properly worn, but the safety harness hook is properly used (NY); 3. The hardhat is properly worn, but the safety harness hook is not used (YN); 4. The work starts directly without the hardhat and the safety harness hook being properly worn (NN). For each situation, 300 videos were recorded. Then, we extracted 3208 images from the videos as a data set, established a YOLO v5 model that can detect the hardhat and the safety harness hook, the mAP is 0.991. The coordinates of the hardhat and safety harness hook were output using the detection model. Using Openpose, an open-source project of Carnegie Mellon University, the coordinates of keypoints 3, 4, 6, 7, 15, 16, 17, and 18 were extracted. Finally, to eliminate the influence of the operator's spatial position on the coordinate values, all the coordinates were integrated into a time series, and the relative values were calculated. A 1D-CNN model is trained using the relative coordinate values of 600 videos, and the actual test is performed using the data of another 600 videos. The final recognition accuracy rate is 0.9467 in the experimental scenario. The recognition accuracy of the standardized use of hardhats and safety harnesses reached 0.9531 and 0.9965, respectively. Compared with the classification algorithms BP neural network and SVM, the accuracy is significantly higher, which illustrates the feasibility of the proposed method.

With the further improvement of the method in the future, it can be used together with safety training to monitor and regulate workers' behavior when using PPE from the perspective of AC in practical applications, to reduce the number of casualties in accidents.

CRediT authorship contribution statement

Jiaqi Li: Writing – original draft, Methodology, Investigation, Validation, Software, Data curation, Conceptualization. **Xuefeng Zhao:** Writing – review & editing, Supervision, Conceptualization, Resources, Funding acquisition. **Guangyi Zhou:** Resources, Supervision. **Min-guan Zhang:** Project administration, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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